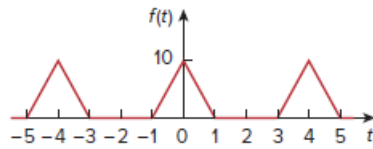


Problem

Determine the trigonometric Fourier series of the signal in Fig. 17.59.

Figure 17.59:



Step-by-step solution

Step 1 of 5

Refer to Equation 17.3 in the textbook.

The Fourier series is,

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

Where,

$$\omega_0 = \frac{2\pi}{T} \text{ is the fundamental frequency.}$$

The function is,

$$f(t) = \begin{cases} -10(t-1), & 0 < t < 1 \\ 0, & 1 < t < 3 \\ 10(t-3), & 3 < t < 4 \end{cases}$$

Step 2 of 5

The time period is, $T = 4$.

Determine the value of ω_0 .

$$\begin{aligned}\omega_0 &= \frac{2\pi}{4} \\ &= \frac{\pi}{2}\end{aligned}$$

Determine the value of a_0 .

$$\begin{aligned}a_0 &= \frac{1}{T} \int_0^T f(t) dt \\ &= \frac{1}{4} \int_0^4 f(t) dt \\ &= \frac{1}{4} \int_0^1 f(t) dt + \frac{1}{4} \int_1^3 f(t) dt + \frac{1}{4} \int_3^4 f(t) dt \\ &= \frac{1}{4} \int_0^1 [-10(t-1)] dt + \frac{1}{4} \int_1^3 0 \cdot dt + \frac{1}{4} \int_3^4 10(t-3) dt \\ &= \frac{10}{4} \int_0^1 [(1-t)] dt + 0 + \frac{10}{4} \int_3^4 (t-3) dt \\ &= \frac{5}{2} \left[(t) - \frac{t^2}{2} \right]_0^1 + \frac{5}{2} \left[\frac{t^2}{2} - 3t \right]_3^4 \\ &= \frac{5}{2} \left[1 - \frac{1}{2} - (0-0) \right] + \frac{5}{2} \left[\frac{4^2}{2} - 3(4) - \left(\frac{3^2}{2} - 3(3) \right) \right] \\ &= \frac{5}{2} \left(\frac{1}{2} \right) + \frac{5}{2} [-4 + 4.5] \\ &= \frac{5}{2}\end{aligned}$$

Step 3 of 5

Determine the value of a_n .

$$\begin{aligned}a_n &= \frac{2}{T} \int_0^T f(t) \cos(n\omega_0 t) dt \\ &= \frac{2}{4} \int_0^4 f(t) \cos(n\omega_0 t) dt \\ &= \frac{1}{2} \int_0^1 f(t) \cos(n\omega_0 t) dt + \frac{1}{2} \int_1^3 f(t) \cos(n\omega_0 t) dt + \frac{1}{2} \int_3^4 f(t) \cos(n\omega_0 t) dt \\ &= \frac{1}{2} \int_0^1 [-10(t-1)] \cos\left(\frac{n\pi}{2} t\right) dt + \frac{1}{2} \int_1^3 0 \cdot \cos\left(\frac{n\pi}{2} t\right) dt + \frac{1}{2} \int_3^4 10(t-3) \cos\left(\frac{n\pi}{2} t\right) dt \\ a_n &= 5 \int_0^1 [-(t-1)] \cos\left(\frac{n\pi}{2} t\right) dt + 0 + 5 \int_3^4 (t-3) \cos\left(\frac{n\pi}{2} t\right) dt\end{aligned}$$

Step 4 of 5

Use the following formula:

$$\int uv = u \int v dx - \int u' \left(\int v dx \right) dx$$

Simplify the equation further,

$$\begin{aligned} a_n &= \left\{ \frac{10}{n\pi} \left[-(t-1) \sin \left(\frac{n\pi}{2} t \right) \right] \right\}_{t=0}^{t=1} - \frac{20}{n^2 \pi^2} \left[\cos \left(\frac{n\pi}{2} t \right) \right]_{t=0}^{t=1} \\ &\quad + \frac{10}{n\pi} \left[(t-3) \sin \left(\frac{n\pi}{2} t \right) \right]_{t=3}^{t=4} + \frac{20}{n^2 \pi^2} \left[\cos \left(\frac{n\pi}{2} t \right) \right]_{t=3}^{t=4} \\ &= \frac{20}{n^2 \pi^2} \left[1 - \cos \left(\frac{n\pi}{2} \right) \right] + \frac{20}{n^2 \pi^2} \left[1 - \cos \left(\frac{3n\pi}{2} \right) \right] \\ &= \frac{20}{n^2 \pi^2} \left[2 - \cos \left(\frac{n\pi}{2} \right) - \cos \left(\frac{3n\pi}{2} \right) \right] \\ &= \frac{20}{n^2 \pi^2} \left[2 - 2 \cos \left(\frac{n\pi}{2} \right) \cos(n\pi) \right] \\ &= \frac{40}{n^2 \pi^2} \left[1 - \cos \left(\frac{n\pi}{2} \right) \right] \end{aligned}$$

Step 5 of 5

Since the function is even, $b_n = 0$.

The Fourier series is,

$$\begin{aligned} f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\ &= \frac{5}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t) + 0 \\ &= \frac{5}{2} + \sum_{n=1}^{\infty} \left\{ \frac{40}{n^2 \pi^2} \left[1 - \cos \left(\frac{n\pi}{2} \right) \right] \cos \left(\frac{n\pi}{2} t \right) \right\} \end{aligned}$$

Thus, the Fourier series of the periodic function is,

$$\boxed{f(t) = \frac{5}{2} + \sum_{n=1}^{\infty} \left\{ \frac{40}{n^2 \pi^2} \left[1 - \cos \left(\frac{n\pi}{2} \right) \right] \cos \left(\frac{n\pi}{2} t \right) \right\}}.$$